🩺 **HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026**

NephroTwin AI

A personalized dialysis digital twin that predicts intradialytic hypotension and supports safer fluid-removal planning.

Predict the risk. Understand the patient. Plan dialysis more safely.

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EXECUTIVE SUMMARY

Problem : Hypotension during dialysis is patient-specific, and population risk scores miss it.

Solution : A persistent virtual patient that scores risk before connection and shows why.

Evidence. AUC 0.977 on a synthetic cohort. On real HEMOBP data the hybrid matches the standard model at 0.871, and we report that openly.

Path. Prospective study, then a SaMD pathway, then machine and EHR partnerships.

Repo: github.com/Shada0/ZWEIKERN_Vel-Tech-Rangarajan-Dr-Sagunthala-RD-Institute
Demo video: https://youtu.be/qpSqBKuTWAU
Live app: https://nephrotwin-ai.streamlit.app/

Simulated research prototype: synthetic cohort plus open real-data validation (HEMOBP). Not a medical device.

🩺 THE CLINICAL PROBLEM

Hypotension during dialysis is patient-specific

Removal outruns refill

Dialysis removes fluid gained between sessions. Removing it faster than the body refills it can drop blood pressure mid-treatment.

Patients differ

Refill rate, vascular response and comorbidity vary sharply from one patient to the next.

Sessions differ

Fluid gain, heat and gap length change the risk for the same patient from one session to the next.

9.5%

of 14,638 real HEMOBP sessions had a hypotensive reading

Population scores ignore each patient’s physiology. **Our target:** the probability of hypotension before the session starts, the reasons behind it, and what-if scenarios for clinicians, never orders.

⚡ THE DIGITAL TWIN SOLUTION

A persistent virtual patient that learns

1

Initialise

Estimate refill rate k, BP sensitivity S and fluid-gain rate g from past sessions, with uncertainty. MAP least squares plus a population prior from training patients only.

2

Update

A scalar Kalman filter fuses home and clinic weights into a fluid-excess state. Each new session re-fits the parameters with a warm start.

3

Simulate

Two-compartment ODE (SciPy), BP response and AR(1) noise. Monte Carlo over uncertainty gives P(SBP < 90 mmHg) for the plan and each what-if.

Outcome label (demo definition): any monitored SBP below 90 mmHg.

Prediction timestamp: scheduled connection time. No later data are used.

📊 DUAL DATA STREAMS

Two streams, controlled for leakage

Stream A · History (EHR)

- Age, sex, diabetes, hypertension, CVD
- Baseline BP, dry weight, labs, medications
- 6 months of sessions: UF, duration, IDWG
- Timed BP and HR, prior hypotension events

Stream B · Dynamic signals

- Daily home weight and BP cuff readings
- Wearable HR, HRV, activity, sleep (gaps shown)
- Temperature and heat-wave flag
- Pre-dialysis vitals

Prediction timestamp

Only data stamped before the scheduled connection time feed any model.

Past sessions only

Twin parameters for session n use sessions before n, never later ones.

Tested, not assumed

Patient-grouped and temporal hold-outs; tests assert outcomes never enter features.

Real HEMOBP data has no wearables, weather or labs. Those streams stay synthetic and are labelled as such.

⚡ SYSTEM ARCHITECTURE

From two data streams to a pre-session risk

Stream A

Historical EHR. Feature extraction uses sessions before the prediction time only.

Stream B

Sensors and weather. Preprocessing adds missingness flags and 3-day windows.

Physiology engine

State. MAP least squares (k, S, g) with shrinkage, plus a Kalman filter for fluid excess.

Fluid. Two-compartment ODE (SciPy).

BP. Blood-volume response with AR(1) noise.

Risk. Monte Carlo over parameter and state uncertainty, then Platt calibration.

Prediction timestamp = scheduled connection time

Models compared

Population baseline (LR, RF)
Personalized digital twin
Hybrid: standard + twin
Ablation: population params

Clinician outputs

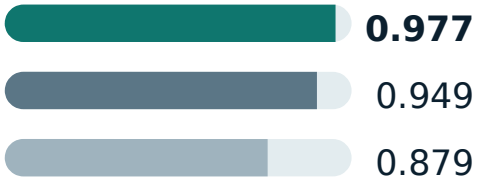
Risk with uncertainty and Green/Amber/Red demo thresholds
What-if simulation
Streamlit dashboard

📊 **EVIDENCE · ROC-AUC, BARS START AT 0.5 (CHANCE)**

Strong on synthetic data, honest on real data

Synthetic cohort

120 patients · 36 held-out test patients · 23.5% prevalence

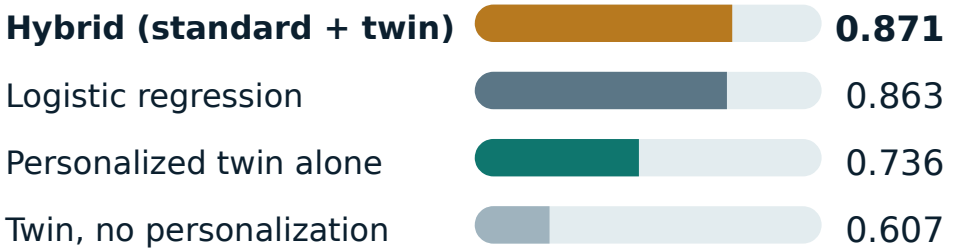


+0.028 AUC vs logistic regression (95% CI +0.015 to +0.043). Personalization lifts the twin from 0.879 to 0.977.

Optimistic: the twin's structure resembles the data generator.

Real open data (HEMOBP)

328 patients · 14,638 sessions · 87 test patients · 9.5% hypotension



Twin alone: −0.127 vs logistic regression (CI −0.159 to −0.097). **Hybrid: +0.008** (CI +0.000 to +0.016), about on par.

Reported openly: sparse readings (median 6 per session) limit identifiability.

🏥 **CLINICIAN DASHBOARD**

13 views and a guided case study

Unit · 4 views

Project overview, unit overview, patient list, weekly priority list (risk × uncertainty, no treatments assigned)

Patient · 2 views

Twin profile with parameter uncertainty and explanations; pre-session card with the what-if planner

Story · 3 views

Ravi guided demo, historical replay (prediction first, outcome after), simulated daily check-in

Trust · 4 views

Model evaluation, real-data validation, data quality, limitations and safety

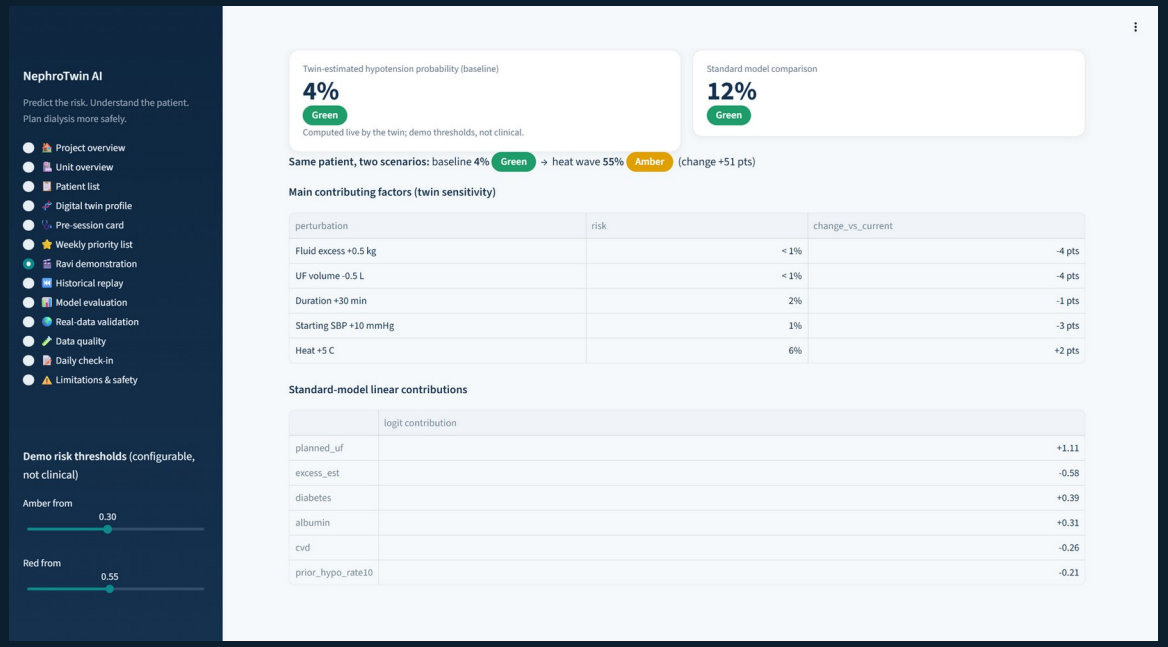
The “Ravi” case · 7 steps

1 Patient overview
2 Dynamic data update
3 Heat-wave scenario
4 Digital twin update
5 Risk flag
6 What-if planner
7 Clinical review

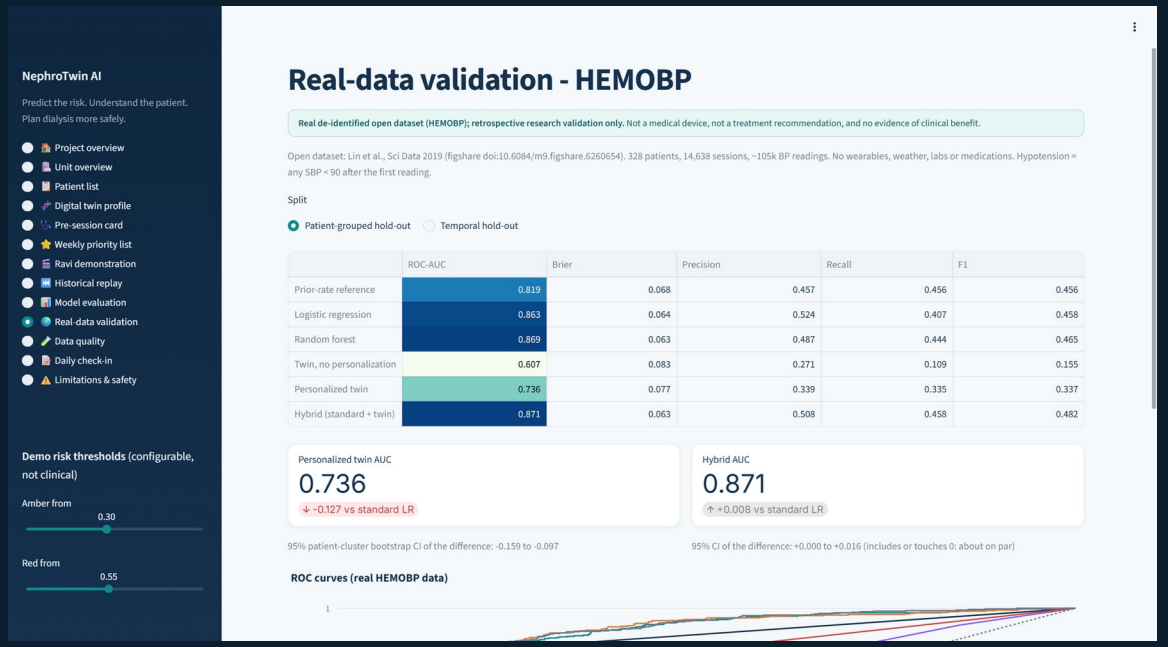
Simulated patient, not real clinical data. The 2.8 kg gain by day 3 is a demonstration input.

DASHBOARD IN ACTION

Ravi: 4% baseline risk becomes 55% in a heat wave



Ravi guided demo, step 5 (live app). The twin recomputes: the same plan scores 4% (Green) at baseline and 55% (Amber) with the synthetic heat-wave input. Synthetic patient, demonstration input.



Real-data validation page (live app). The hybrid edges the standard model (0.871 vs 0.863) while adding physiology and what-if simulation; the twin alone trails. We report that openly.

💼 BUSINESS VALUE AND CLINIC ECONOMICS

Value, pricing and ROI for dialysis clinics

🏥 Value for chains and hospitals

- Fewer emergency intradialytic interventions
- Shorter nurse intervention times
- Fewer early session terminations
- Better patient retention and survival

Targets for the prospective study, not yet demonstrated.

💼 Pricing model

PPPM licence. Per-patient-per-month software fee for outpatient dialysis centers: **$25 PPPM (US/EU); India tier about $10**

EHR integration fee. Enterprise fee per site for Epic or Cerner connections: **$10,000 one-time per site**

Priced against avoided events: cost per acute event prevented, and per avoided emergency admission.

💵 ROI per clinic, per year

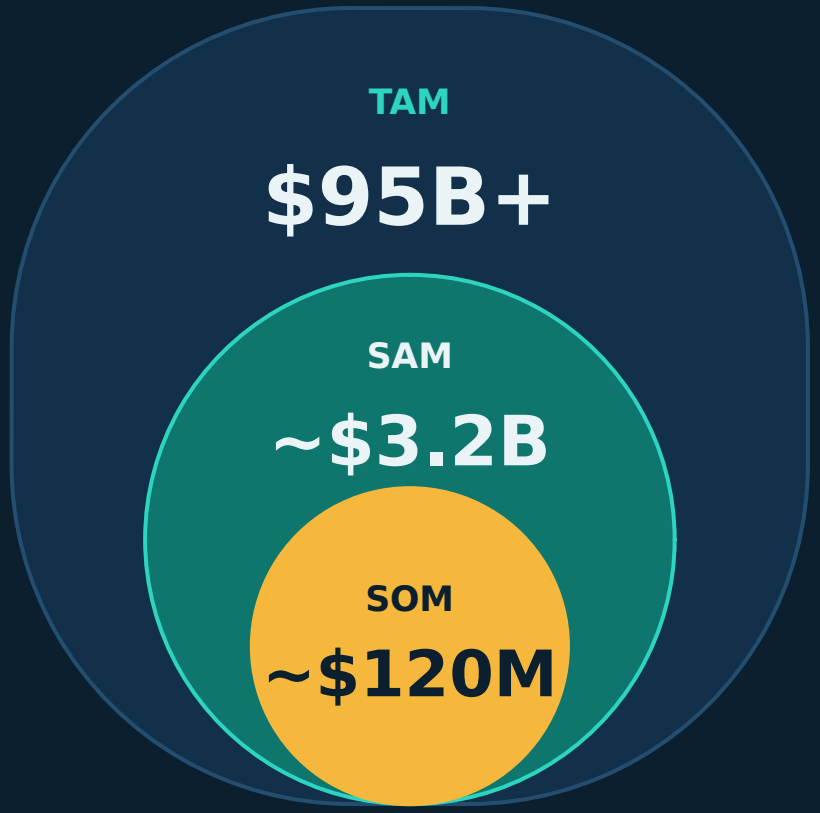
+ Avoided acute events × $100
+ Avoided hospitalizations × $4,000
+ Nurse time saved × $25
− Licence: patients × 12 × PPPM

Planning assumptions, not results. No savings are claimed until the Phase 1 study.

Worked example (illustrative planning model): 100-patient centre, 15,600 sessions a year, 10% needing an acute intervention, 15% target reduction = 234 events avoided. Benefit about $39.8K (events $23.4K + nurse time $4.4K + 3 avoided emergency admissions $12.0K) against a $30.0K licence: net about +$9.8K, benefit-to-cost 1.3x. Cost per acute event prevented (licence ÷ events) is about $128; each avoided emergency admission is valued at $4,000. Break-even needs roughly 10% fewer interventions. All inputs get replaced by Phase 1 measurements.

📊 MARKET OPPORTUNITY

From a $95B+ market to a focused first wedge



TAM · Global dialysis

About $95B+ global dialysis market serving 3.8M+ end-stage renal disease patients worldwide.

SAM · Clinical decision support

Software and AI decision support in North American and European dialysis clinics, about $3.2B.

SOM · Initial rollout

Top 50 regional outpatient dialysis chains, about $120M addressable.

India lens: about 2.2 lakh new ESRD patients a year (roughly 3.4 crore added dialysis sessions). India-first entry via private chains and district centres.

Global figures: team triangulation of published market-research estimates; verify before external use. India figures: Ministry of Health and Family Welfare, PM National Dialysis Programme guidelines.

🚀 GO-TO-MARKET AND REGULATORY ROADMAP

Validate, regulate, then scale

PHASE 1 · NEAR TERM · 0 TO 12 MONTHS

🔬 Validate

- IRB-approved prospective observational study with partner clinics
- Continuous BP, medications and UF profiles
- Real-world value of continuous monitoring
- Clinician usability study

PHASE 2 · REGULATORY · 12 TO 30 MONTHS

🛡️ Regulate

- Software as a Medical Device (SaMD) as a clinical decision support tool
- FDA 510(k) submission and CE mark pathway
- Quality system, clinical validation, calibration monitoring

Pathway to confirm with regulatory counsel. India CDSCO pilots run in parallel.

PHASE 3 · SCALE · 30 TO 48 MONTHS

🏥 Scale

- Integration partnerships with dialysis machine makers, e.g. Fresenius, Baxter
- EHR integrations, e.g. Epic, Cerner
- Rollout across the top 50 regional chains

Target partners, no agreements yet.

Commercial motion. Pilot with partner clinics, convert to PPPM licences, add EHR integration fees as sites connect.

COMPETITIVE LANDSCAPE

Twin versus standard ML versus static rules

	Static rules	Standard ML	NephroTwin AI
Personalization	None. Fixed thresholds.	Learned from population features.	**Per-patient k, S, g,** updated every session.
Explainability	Transparent but generic.	Feature contributions.	**Physiology parameters** and fluid state.
What-if simulation	Not supported.	Re-scores inputs, no physiology.	**Simulates** duration, UF, gap and heat.
Uncertainty	None.	Probability only.	**Parameter and state** uncertainty.

Honest edge. On real data the hybrid matches the standard model (0.871 vs 0.863) and adds explainable physiology and what-if simulation.

SAFETY, GOVERNANCE AND HONEST REPORTING

Built to inform clinicians, never to instruct them

Prototype status

A simulated research prototype, not a medical device. Synthetic patients are labelled everywhere. Thresholds are configurable demonstration values.

Non-directive by design

Planner outputs are simulations marked “Clinical review required” and “Not a treatment recommendation”. No orders for duration, UF, medication or sessions.

Honest reporting

On real data the twin alone underperforms. Synthetic wins are not clinical effectiveness, and retrospective simulation cannot show treatment benefit.

Required before any real use

- Clinical validation
- Regulatory review
- Data-protection review
- Expert oversight

🚀 FUTURE SCOPE AND CALL TO ACTION

From a research prototype to a clinical tool

Now

Prospective, ethics-approved validation with richer data: medications, continuous BP, UF profiles.

Next

Hierarchical Bayesian pooling and a better observation model for sparse readings. Clinician usability study. Real wearable and weather feeds.

Later

Calibration monitoring, the SaMD submission, and machine and EHR partnerships.

Partner with us

We seek **clinic partners** for the Phase 1 study, **nephrology advisors**, and **seed funding for the Phase 1 study**.

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